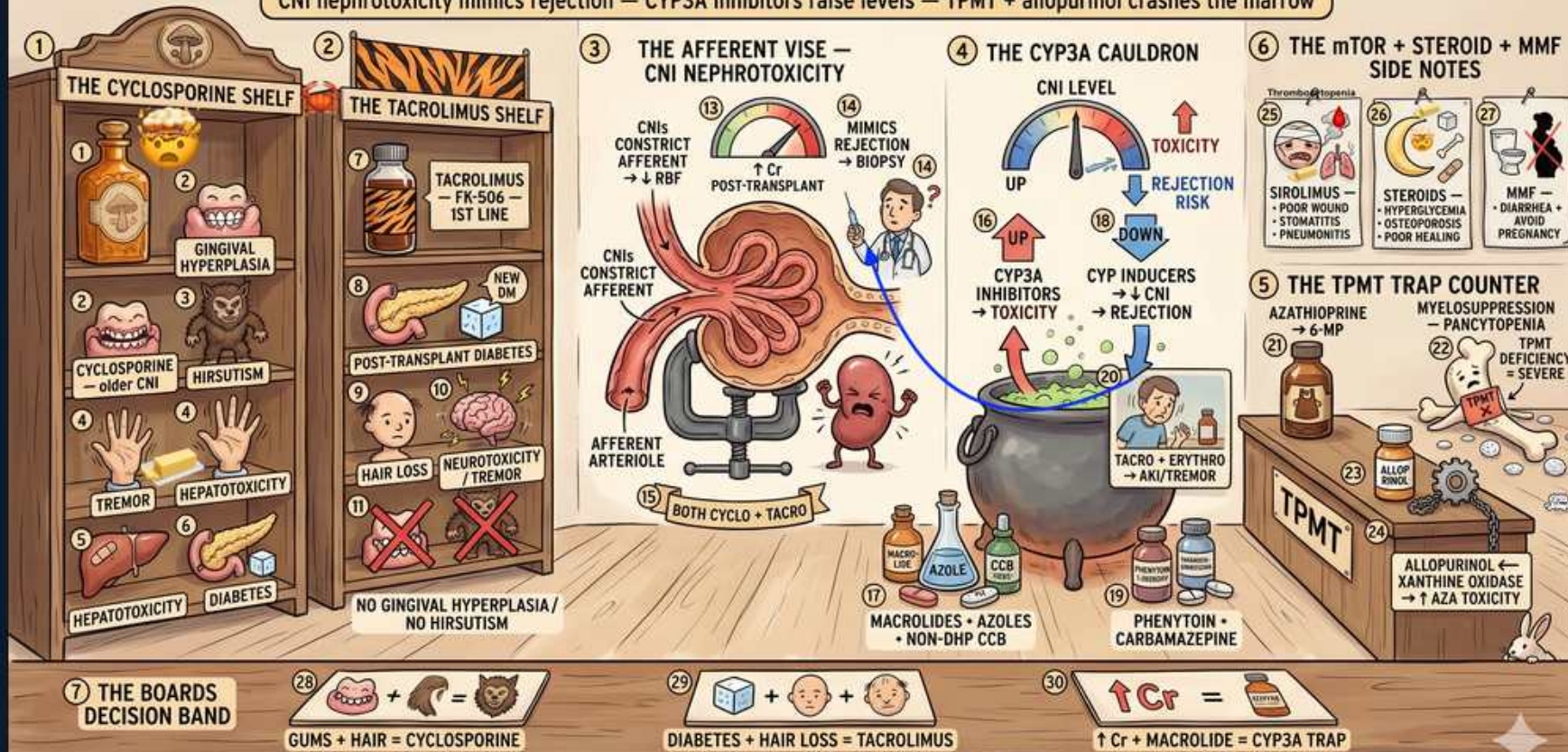


Transplant Drug Toxicities, Nephrotoxicity & CYP3A Interactions

THE TOXICITY APOTHECARY — TRANSPLANT DRUG SIDE EFFECTS, NEPHROTOXICITY & CYP3A INTERACTIONS

CNI nephrotoxicity mimics rejection — CYP3A inhibitors raise levels — TPMT + allopurinol crashes the marrow



Both CNIs are nephrotoxic via afferent constriction. CYP3A inhibitors raise levels — toxicity. TPMT + allopurinol with azathioprine — pancytopenia.

1. Cyclosporine — gingival hyperplasia, hirsutism

WHAT YOU SEE

Cyclosporine shelf: gums, hair, tremor, hepatotoxicity

WHAT IT ENCODES

Cyclosporine (older CNI) classically causes gingival hyperplasia, hirsutism, tremor, hepatotoxicity, and hyperuricemia/gout. It shares nephrotoxicity with tacrolimus via afferent vasoconstriction.

BOARD TRAP

Gingival hyperplasia + hirsutism point to cyclosporine — NOT tacrolimus, which causes hair loss and diabetes instead.

2. Tacrolimus — diabetes, hair loss, neurotoxicity

WHAT YOU SEE

Tacrolimus shelf: post-transplant diabetes, hair loss, tremor/neurotox

WHAT IT ENCODES

Tacrolimus (FK506, first-line CNI) causes new-onset diabetes after transplant, alopecia, tremor/neurotoxicity, and nephrotoxicity. NO gingival hyperplasia or hirsutism (the cyclosporine discriminators).

BOARD TRAP

Tacrolimus → diabetes + hair LOSS; if the stem says gum overgrowth/hirsutism, the drug is cyclosporine, not tacrolimus.

3. CNI nephrotoxicity mimics rejection

WHAT YOU SEE

Afferent vise, CNI constrict afferent, mimics rejection

WHAT IT ENCODES

Both CNIs cause dose-dependent nephrotoxicity by constricting the AFFERENT arteriole, dropping GFR. On biopsy/clinically it can look like rejection — check trough levels to distinguish.

BOARD TRAP

A rising creatinine on a CNI may be drug nephrotoxicity (lower the dose), NOT rejection (raise immunosuppression) — biopsy + level decide.

4. Afferent arteriole constriction → low GFR

WHAT YOU SEE

Afferent arteriole diagram, vasoconstriction

WHAT IT ENCODES

CNI-induced afferent vasoconstriction reduces renal blood flow and GFR, raising creatinine acutely and causing chronic interstitial fibrosis with prolonged exposure.

BOARD TRAP

CNI nephrotoxicity is afferent (pre-glomerular) — distinct from NSAID afferent effect plus ACEi efferent dilation; the fix is dose reduction.

5. CYP3A4 — central metabolism of CNIs

WHAT YOU SEE

CYP3A cauldron, level UP = toxicity, DOWN = rejection

WHAT IT ENCODES

CNIs and mTOR inhibitors are metabolized by CYP3A4. Anything that changes CYP3A activity swings drug levels: inhibitors raise levels (toxicity), inducers lower levels (rejection risk).

BOARD TRAP

CYP3A inhibitors RAISE CNI levels → nephrotoxicity; inducers LOWER levels → under-immunosuppression and rejection — opposite directions.

6. CYP3A inhibitors — azoles, macrolides, non-DHP CCB

WHAT YOU SEE

Inhibitors list: azoles, macrolides, diltiazem/verapamil

WHAT IT ENCODES

Azole antifungals (fluconazole/voriconazole), macrolides (erythro/clarithromycin), and non-dihydropyridine CCBs (diltiazem, verapamil) inhibit CYP3A4 → elevated CNI levels → toxicity. Reduce CNI dose.

BOARD TRAP

Adding an azole or clarithromycin to a tacrolimus patient spikes the level — anticipate toxicity and pre-emptively cut the CNI dose.

7. CYP3A inducers — rifampin, phenytoin, carbamazepine

WHAT YOU SEE

Inducers: rifampin, phenytoin, carbamazepine

WHAT IT ENCODES

Rifampin, phenytoin, and carbamazepine induce CYP3A4 → subtherapeutic CNI levels → acute rejection risk. Anticipate the need to raise CNI dose and monitor levels closely.

BOARD TRAP

Rifampin can precipitate rejection by dropping tacrolimus levels — the answer is increase CNI dosing/monitor, not 'no interaction.'

8. mTOR inhibitor side effects

WHAT YOU SEE

mTOR + steroid + MMF side-notes panel

WHAT IT ENCODES

Sirolimus/everolimus cause poor wound healing, proteinuria, hyperlipidemia, mouth ulcers, and pneumonitis. They are non-nephrotoxic but should be avoided peri-operatively.

BOARD TRAP

Sirolimus = impaired wound healing + proteinuria; don't blame proteinuria solely on recurrent disease in a sirolimus patient.

9. Steroid side effects

WHAT YOU SEE

Steroid notes: diabetes, osteoporosis, hypertension

WHAT IT ENCODES

Chronic glucocorticoids cause hyperglycemia, osteoporosis/AVN, hypertension, weight gain, and infection risk — compounding tacrolimus's diabetogenic effect.

BOARD TRAP

Post-transplant hyperglycemia is multifactorial — steroids AND tacrolimus both contribute; both may need adjustment.

10. MMF side effects — GI & marrow

WHAT YOU SEE

MMF notes: diarrhea, leukopenia

WHAT IT ENCODES

Mycophenolate causes dose-related GI symptoms (diarrhea, nausea) and marrow suppression (leukopenia/anemia). It is teratogenic. Differentiate GI infection from MMF GI toxicity.

BOARD TRAP

New diarrhea on MMF may be drug toxicity OR CMV colitis — work up CMV before just reducing MMF in a sick patient.

11. TPMT trap — azathioprine myelosuppression

WHAT YOU SEE

TPMT trap counter, azathioprine/6-MP marrow

WHAT IT ENCODES

Azathioprine is inactivated by TPMT; TPMT-deficient patients accumulate active metabolite → severe pancytopenia. Check TPMT before/at dosing; this is independent of CYP3A.

BOARD TRAP

Azathioprine toxicity is a TPMT (not CYP3A) issue — pancytopenia in a TPMT-deficient patient, worsened further by allopurinol.

12. Allopurinol + azathioprine → pancytopenia

WHAT YOU SEE

Allopurinol blocks xanthine oxidase → AZA toxicity

WHAT IT ENCODES

Allopurinol inhibits xanthine oxidase, the enzyme that clears azathioprine's active metabolite, causing dangerous accumulation and pancytopenia. Avoid the combination or drastically reduce AZA.

BOARD TRAP

Allopurinol + azathioprine is the classic lethal interaction — pancytopenia; this is xanthine oxidase, NOT a CYP3A effect.

13. Boards decision band

WHAT YOU SEE

Bottom band: gums/hair=cyclosporine, diabetes=tacrolimus, ↑Cr+macrolide=CYP3A

WHAT IT ENCODES

High-yield: gingival hyperplasia/hirsutism = cyclosporine; diabetes/hair loss = tacrolimus; rising Cr + azole/macrolide = CYP3A inhibition → CNI toxicity; allopurinol + azathioprine = pancytopenia.

BOARD TRAP

The two killer interactions tested: CYP3A inhibitors → CNI toxicity, and allopurinol → azathioprine pancytopenia — keep the enzymes straight.